Classical Field Theory — Exercise Sheet 22

Vortex orientation, scalar reduction and moduli

Solutions

Throughout this sheet $q = v = \lambda = 1$, $D_j = \partial_j - \mathrm{i}A_j$, $B = \partial_1 A_2 - \partial_2 A_1$, and the plane has orientation $\mathrm{d}x_1 \wedge \mathrm{d}x_2$. Use Lecture 22's energy and current $K_j = \frac{1}{2}\operatorname{Im}(\overline{\Phi} D_j \Phi)$. Every sector statement retains its finite-energy, uniform vacuum-modulus and absolute magnetic-integrability assumptions.

Exercise 22.1 (Reflection and higher-winding antivortices). Let $R(x_1, x_2) = (x_1, -x_2)$. For smooth fields set $\widetilde{\Phi} = \Phi \circ R$ and $\widetilde{A} = R^* A$.

(a) Track B, K, the energy and counterclockwise winding under this reflection. Derive the square completion for winding $-N$, $N \geq 1$, and prove the bound and equality equations in the reflected sector.

(b) For real radial profiles use $\Phi = f(r)e^{-\mathrm{i}N\theta}$ and $A = \alpha(r)\mathrm{d}\theta$. Derive both equality ODEs, the flux and the boundary conditions on the positive vacuum branch. Give sufficient smooth Cartesian core forms and the first corrections to $f = s_N r^N + \cdots$ and $\alpha = -r^2/4 + \cdots$. Here $s_N > 0$ is a local slope coefficient, not a proved global shooting value.

Solution. The pulled-back components are $\widetilde{A}_1 = A_1 \circ R$ and $\widetilde{A}_2 = -A_2 \circ R$. Thus $\widetilde{B} = -B \circ R$. The covariant derivatives obey $D_1\widetilde{\Phi} = (D_1\Phi)\circ R$ and $D_2\widetilde{\Phi} = -(D_2\Phi)\circ R$. Their squared norms, the magnetic square and scalar potential therefore give the same energy after the change of variables. The current components transform as a one-form, giving $\operatorname{curl}\widetilde{K} = -(\operatorname{curl} K)\circ R$. A counterclockwise circle becomes clockwise, so winding changes sign.

Apply the positive completion of Lecture 22 to the unreflected fields and express it through the reflected ones. Dropping tildes gives

$$e_1 = \frac{1}{2}|(D_1 - \mathrm{i}D_2)\Phi|^2 + \frac{1}{2}\left[B + \frac{1 - |\Phi|^2}{2}\right]^2 - \frac{B}{2} - \operatorname{curl} K.$$

The reflected sector has flux $-2\pi N$. Reflection preserves all absolute-integrability and modulus assumptions. It also transports the lecture's sequence of circles with vanishing current integral. Integration of this identity along that sequence yields

$$E_1 = \pi N + \frac{1}{2}\int_{\mathbb{R}^2}\left[|(D_1 - \mathrm{i}D_2)\Phi|^2 + \left(B + \frac{1 - |\Phi|^2}{2}\right)^2\right]\mathrm{d}^2x.$$

The squares prove the bound and vanish exactly for $(D_1 - \mathrm{i}D_2)\Phi = 0$, $B = -(1 - |\Phi|^2)/2$.

The polar derivative formula, including the angular potential, gives

$$(D_1 - \mathrm{i}D_2)\Phi = e^{-\mathrm{i}(N+1)\theta}\left[f' - \frac{N+\alpha}{r}f\right], \qquad B = \frac{\alpha'}{r}.$$

Hence the radial equations are

$$f' = \frac{N+\alpha}{r}f, \qquad \frac{\alpha'}{r} = -\frac{1}{2}(1 - f^2).$$

Sufficient smooth core forms are $f = r^N F(r^2)$ and $\alpha = r^2 G(r^2)$, with smooth real F, G. Indeed, $\Phi = (x_1 - \mathrm{i}x_2)^N F(r^2)$ and $A = G(r^2)(-x_2 \mathrm{d}x_1 + x_1 \mathrm{d}x_2)$ are smooth. They imply $f(0) = \alpha(0) = 0$. Choose $f \to 1$ at infinity. Radial flux is $2\pi[\alpha(\infty) - \alpha(0)] = -2\pi N$, so $\alpha \to -N$.

The second equation first gives $\alpha = -r^2/4 + O(r^{2N+2})$. The first then gives $(\log(f/r^N))' = \alpha/r = -r/4 + O(r^{2N+1})$. Thus $f = s_N r^N(1 - r^2/8 + O(r^4))$. Substituting its square in $\alpha' = -r/2 + rf^2/2$ gives

$$\alpha = -\frac{r^2}{4} + \frac{s_N^2}{4N+4} r^{2N+2} + O(r^{2N+4}).$$

This is a local necessary expansion of a smooth radial equality field. It does not establish a global solution for every s_N.

Exercise 22.2 (The scalar equation with its point sources). Let a positive critical vortex in the stated sector have finitely many zeros $z_j \in \mathbb{R}^2$ of multiplicities $m_j \geq 1$. Away from the zeros put $w = \log|\Phi|^2$. Assume as local zero data that $w = 2m_j \log|x - z_j| + w_j$ near z_j, where w_j is smooth. This local regular remainder is a hypothesis, not a claim about arbitrary smooth fields with zeros. Use the distributional identity $\Delta \log|x - z_j| = 2\pi\delta_{z_j}$ as a given.

(a) On a phase patch derive A_1, A_2 in terms of w and the real phase. Eliminate the potential from the two Bogomolny equations.

(b) Prove the resulting equation on the whole plane as a distribution, including its point-source coefficient. Determine $\int_{\mathbb{R}^2}(1 - e^w)\mathrm{d}^2x$ from the sector flux when $\sum_j m_j = N$. Does this reduction prove existence for prescribed zeros?

Solution. On a phase patch the real and imaginary parts of the first equality equation give

$$A_1 = \partial_1\theta + \frac{1}{2}\partial_2 w, \qquad A_2 = \partial_2\theta - \frac{1}{2}\partial_1 w.$$

Mixed phase derivatives cancel, leaving $B = -\Delta w/2$ away from the zeros. The second equality equation is $B = (1 - e^w)/2$, so $\Delta w + 1 - e^w = 0$ there.

Near z_j, the given logarithmic part contributes $4\pi m_j \delta_{z_j}$ to the distributional Laplacian. The function $e^w = |x - z_j|^{2m_j} e^{w_j}$ is smooth across that point. The regular function $\Delta w_j + 1 - e^w$ vanishes on the punctured neighbourhood, so by continuity it vanishes throughout it. Combining these local statements with the equation off the zero set gives

$$\Delta w + 1 - e^w = 4\pi \sum_j m_j \delta_{z_j}$$

as distributions on the plane. The coefficient is $4\pi m_j$, not $2\pi m_j$, because the logarithm is that of the squared modulus.

Since $1 - e^w = 2B$ extends smoothly and B is absolutely integrable, the integral exists and the positive sector flux gives

$$\int_{\mathbb{R}^2}(1 - e^w)\mathrm{d}^2x = 2\int_{\mathbb{R}^2} B\,\mathrm{d}^2x = 4\pi N.$$

No uncontrolled integral of a Laplacian at infinity was needed. The calculation starts with an existing equality field and prescribed local zero behaviour. Solving the singular nonlinear equation, with its asymptotics and reconstructing a smooth potential, requires the separately cited existence theory.

Exercise 22.3 (Parameters at a two-vortex collision). A *moduli space* here is the set of critical positive N-vortex solutions with the vacuum asymptotics, modulo smooth gauge transformations. Use the following stronger geometric result as a *given*: this space is a nonsingular complex manifold, and the coefficients of

$$P(z) = \prod_{j=1}^{N} (z - z_j)$$

are smooth complex coordinates, including at coincident zeros. This input is not inferred from existence and uniqueness as a set alone. For its zero-coordinate description see Alonso Izquierdo, Garcia Fuertes, Manton and Mateos Guilarte, *JHEP* **01** (2024) 020, §5, pp. 10–11, equation (5.1) and discussion. Taubes's existence/uniqueness result is the one cited in Lecture 22.

(a) Count the real moduli and the moduli after fixing the mean zero position. Explain why ordering the zeros overcounts configurations.

(b) For $N = 2$, write $P(z) = z^2 - sz + p$. Replace (s, p) by the centre $Z = s/2$ and discriminant $\Delta = s^2 - 4p$. Locate the collision set and determine its real codimension. Explain why (z_1, z_2) and their separation are poor smooth coordinates there.

Solution. The leading coefficient of P is fixed to one, leaving N independent complex coefficients. The given coordinate theorem makes their space real $2N$-dimensional. Permuting roots does not change the polynomial or the corresponding gauge class, so labelled roots are redundant. The coefficient of z^{N-1} is $-\sum_j z_j$. Fixing the mean position to zero sets this one complex coefficient to zero and leaves $2N - 2$ real parameters.

For $N = 2$, put $Z = s/2$ and $\Delta = s^2 - 4p$. The inverse transformation is $s = 2Z$, $p = Z^2 - \Delta/4$, so it is a smooth invertible change of complex coordinates. It gives

$$P(z) = (z - Z)^2 - \frac{\Delta}{4}, \qquad z_{1,2} = Z \pm \frac{\sqrt{\Delta}}{2}.$$

The collision occurs exactly when $\Delta = 0$. The coordinate Z remains arbitrary, giving a complex one-dimensional, real two-dimensional subset of the real four-dimensional space. The ambient coefficient chart is nonsingular there.

The root-to-coefficient Jacobian has determinant $z_1 - z_2$ up to sign, so it degenerates at a collision. The inverse requires a square root and a choice of root labels. For centred roots at $\pm d$, one has $\Delta = 4d^2$. Thus d, or the nonnegative distance $2|d|$, is not a smooth coordinate at the collision even though Δ is. These are consequences of the supplied smooth-coordinate input and polynomial algebra, not a proof of that analytic moduli-space input.

Exercise 22.4 (Designing a radial computation). For the positive unit-vortex ODEs $f' = (1 - \alpha)f/r$ and $\alpha'/r = (1 - f^2)/2$, design a computation on $[r_0, R]$, where $0 < r_0 \ll 1 < R$.

(a) Derive the coefficients through r^5 in f and through r^6 in α, using the free coefficient $s_0 = f'(0)$.

(b) Treat s_0 as an unknown parameter. Give three boundary residuals for a two-equation boundary-value solver using the core series and $f(R) = 1$. Explain why adding $\alpha(R) = 1$ independently would overdetermine this particular setup.

(c) Specify separate tests of origin truncation, outer-boundary truncation and solver accuracy. What further information is needed before a numerical residual could prove existence of a full-plane vortex? No numerical run is required in this problem.

Solution. Write $f = s_0 r + c_3 r^3 + c_5 r^5 + \cdots$ and $\alpha = a_2 r^2 + a_4 r^4 + a_6 r^6 + \cdots$. The magnetic

equation gives

$$2a_2 = \frac{1}{2}, \qquad 4a_4 = -\frac{s_0^2}{2}, \qquad 6a_6 = -s_0 c_3.$$

The first equation gives $3c_3 = c_3 - a_2 s_0$ and $5c_5 = c_5 - a_2 c_3 - a_4 s_0$. Therefore

$$a_2 = \frac{1}{4}, \quad c_3 = -\frac{s_0}{8}, \quad a_4 = -\frac{s_0^2}{8}, \quad a_6 = \frac{s_0^2}{48}, \quad c_5 = \frac{s_0}{128} + \frac{s_0^3}{32}.$$

Let $f_{\mathrm{core}}(r; s_0)$ and $\alpha_{\mathrm{core}}(r; s_0)$ denote these truncated polynomials. The three endpoint residuals are

$$f(r_0) - f_{\mathrm{core}}(r_0; s_0), \qquad \alpha(r_0) - \alpha_{\mathrm{core}}(r_0; s_0), \qquad f(R) - 1.$$

The two first-order equations contribute two integration constants and the unknown s_0 contributes one parameter. Three scalar residuals match that count. Adding an independent fourth one generally overdetermines it. The computed value of $\alpha(R) - 1$ is instead a useful diagnostic.

To test origin truncation, decrease r_0 while retaining small solver error and fixed R, or increase the core-series order. To test outer truncation, increase R and compare the fitted s_0, the profiles on a common interior interval and the tail diagnostics. To test the solver, refine its mesh/tolerance at fixed endpoints and measure both ODE residuals and endpoint residuals independently. A small collocation residual is not the same as a proved uniform solution error, and none of these finite tests controls the entire infinite tail. A rigorous numerical existence proof would additionally require validated error bounds and control of the singular-origin and infinite-boundary limits. The cited Taubes theorem supplies existence by a separate argument.